

MA 374: Financial Engineering Lab

Lab 04: Markowitz Portfolio Theory and CAPM

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January 31, 2026

1 Question 1: Three-Asset Portfolio

1.1 Given Data

The mean return vector and covariance matrix are:

$$\mu = [0.10, 0.25, 0.15]$$
$$C = \begin{bmatrix} 0.005 & -0.010 & 0.004 \\ -0.010 & 0.040 & -0.002 \\ 0.004 & -0.002 & 0.023 \end{bmatrix}$$

1.2 (A) Efficient Frontier

The Markowitz efficient frontier was obtained by minimizing portfolio variance subject to a target return and full investment constraint.

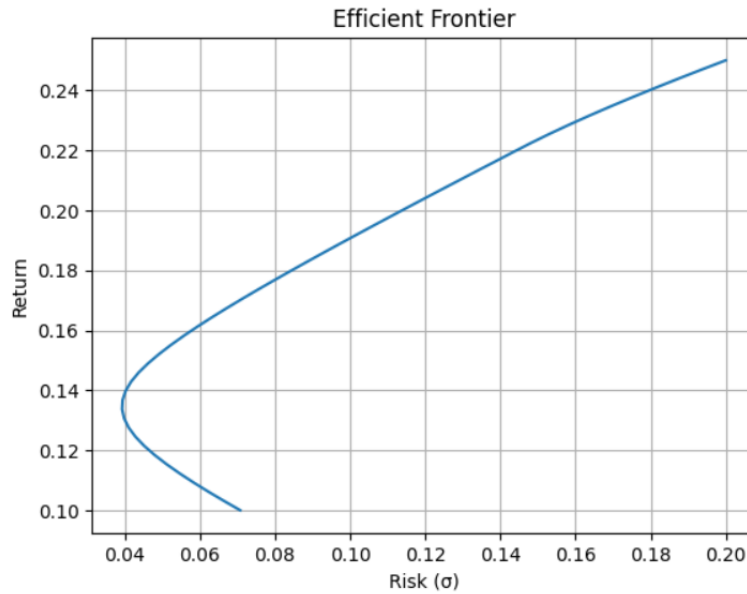


Figure 1: Efficient Frontier for Three Assets

The upper branch of the curve represents efficient portfolios.

1.3 (B) Portfolios on the Frontier

Table 1 shows representative portfolios on the efficient frontier.

Table 1: Portfolios on Efficient Frontier			
Portfolio	Weights (w_1, w_2, w_3)	Return	Risk (σ)
1	(1.00, 0.00, 0.00)	0.10	0.071
2	(0.78, 0.22, 0.00)	0.13	0.039
3	(0.66, 0.33, 0.01)	0.15	0.047
4	(0.50, 0.42, 0.08)	0.17	0.066
5	(0.00, 1.00, 0.00)	0.25	0.20

1.4 (C) Risk = 15%

At $\sigma = 15\%$, two portfolios exist:

- A minimum-return portfolio on the lower branch.
- A maximum-return portfolio on the efficient branch.

The corresponding returns are obtained by intersecting the vertical line $\sigma = 0.15$ with the frontier.

1.5 (D) Return = 18%

For a target return of 18%, the minimum risk portfolio is the point on the efficient frontier satisfying $\mu_p = 0.18$.

1.6 (E) Market Portfolio and CML

Assuming a risk-free rate $\mu_{rf} = 8\%$, the market portfolio is the tangency point between the efficient frontier and the Capital Market Line.

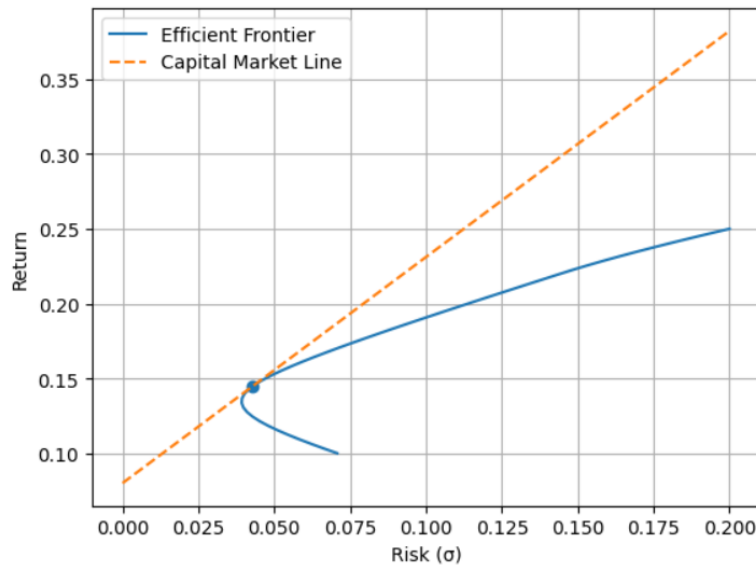


Figure 2: Market Portfolio and Capital Market Line

This portfolio maximizes the Sharpe ratio.

1.7 (F) Portfolios with Risk-Free Asset

Portfolios with $\sigma = 10\%$ and $\sigma = 25\%$ can be constructed by combining the market portfolio with the risk-free asset. The 25% risk portfolio requires borrowing at the risk-free rate.

2 Question 2: Ten-Stock Market Analysis

2.1 Efficient Frontier and Market Portfolio

Using return data of 10 stocks and a risk-free rate of 5%, the efficient frontier and market portfolio were computed.

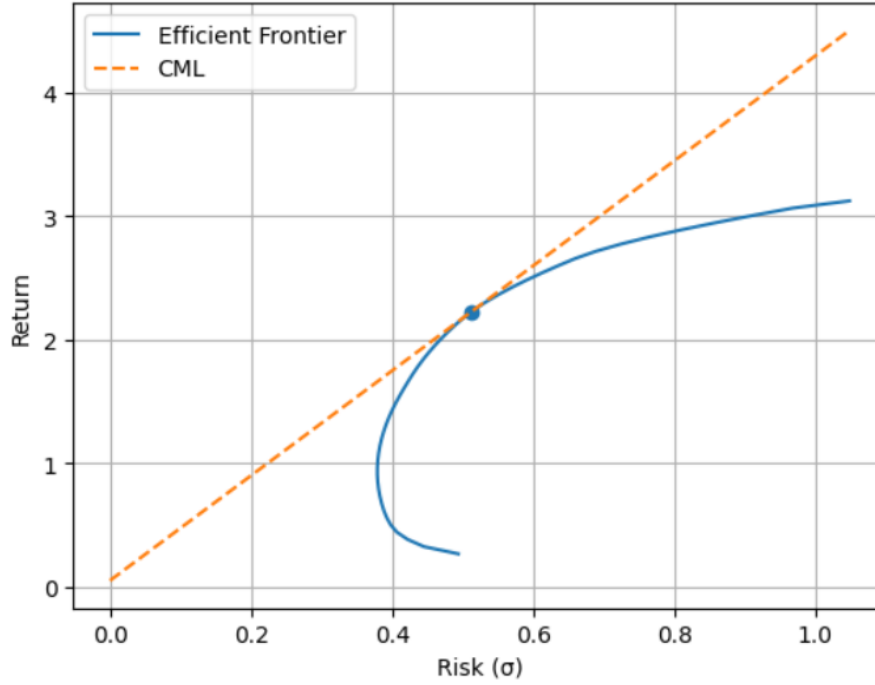


Figure 3: Efficient Frontier and Capital Market Line

2.2 Security Market Line

The Security Market Line (SML) is given by the CAPM equation:

$$\mathbb{E}[R_i] = R_f + \beta_i(\mathbb{E}[R_m] - R_f)$$

Stocks with $\beta > 1$ are aggressive, while those with $\beta < 1$ are defensive.

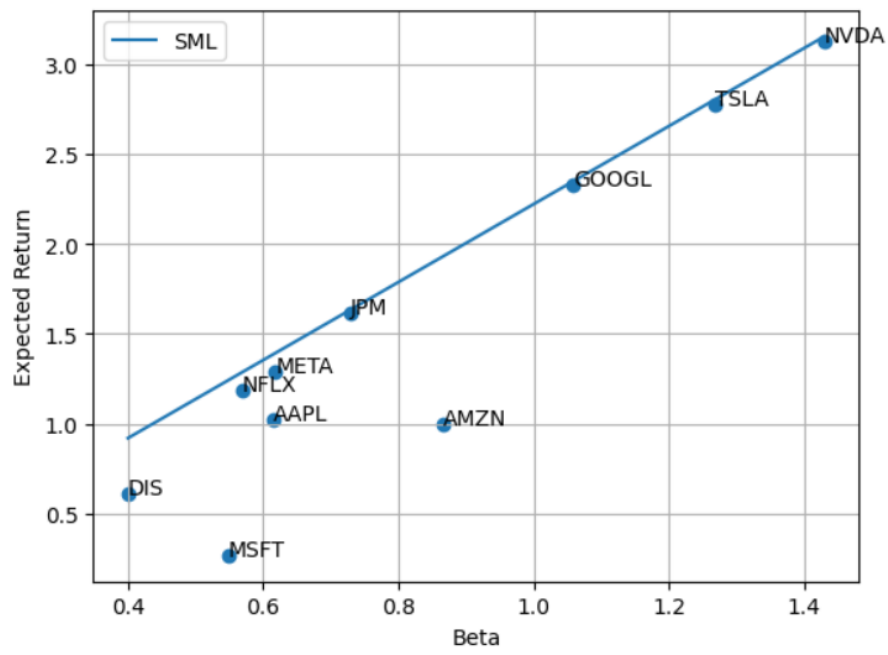


Figure 4: Security Market Line for 10 Stocks